Ramdeobaba University, Nagpur

IPR CELL

PATENT REQUISITES DRAFT

1. **Full Name and Address of the inventor(s) and applicant(s)**

SN	Name of applicant/Inventor	Designation Faculty /UG/PG student/Research scholar	E mail ID	Mobile Number	Address
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2. **What is the suggested title of the invention?**

THIRD EYE FOR THE BLIND - Advanced Wearable Technology Enhancing Mobility for the Visually Impaired

3. **The relevant prior art/existing technology and its disadvantages.**

Existing assistive technologies for visually impaired individuals often rely on traditional tools such as canes or advanced electronic devices like wearable haptic systems. However, these systems may have limitations, including:

- **Limited range**: Many do not provide feedback for distant obstacles or require constant physical interaction (like a cane).
- **Complex interfaces**: Some devices are hard to operate, requiring users to learn intricate systems.
- **High cost and maintenance**: Advanced wearables tend to be expensive and require specialized repairs or replacements.

4. **What are the objectives/advantages of your invention? Technical Problem or problems solved by your invention. Provide a clock diagram.**

The invention, a wearable device in the form of glasses, addresses the limitations of existing technologies by:

- **Providing real-time obstacle detection** using ultrasonic sensors and an auditory buzzer, ensuring an immediate and easy-to-understand response.
- **Simplifying the user interface**: The device only requires a button to power on and off, making it highly accessible for all age groups.
- **Offering a cost-effective and scalable solution**, with a design that is easy to manufacture and maintain.

The invention solves the technical problem of limited obstacle detection and complex interfaces by integrating real-time sensors and a simple haptic feedback system.

5. **Complete description of the invention supported with line diagrams. The description should include all the elements of the invention there construction with respect to other elements and function thereof.** ***(List all parts of your invention or steps in the process if you have a process and how the parts or steps relate to each other).***

The invention is a wearable obstacle detection system designed for visually impaired users. Its key components include:

- **Arduino Nano (ATmega328)**: The central microcontroller that processes data from the sensors and controls the buzzer.
- **Ultrasonic Sensor (HC-SR04)**: Mounted at the front of the glasses, it continuously measures the distance to objects ahead and sends this information to the Arduino.
- **Buzzer**: Provides an auditory alert to the user when an obstacle is detected within a preset distance (e.g., 100 cm).
- **9V Battery**: Powers the entire system, with a design optimized for energy efficiency to ensure extended use.
- **Push Button**: Allows the user to activate or deactivate the device with a single press.

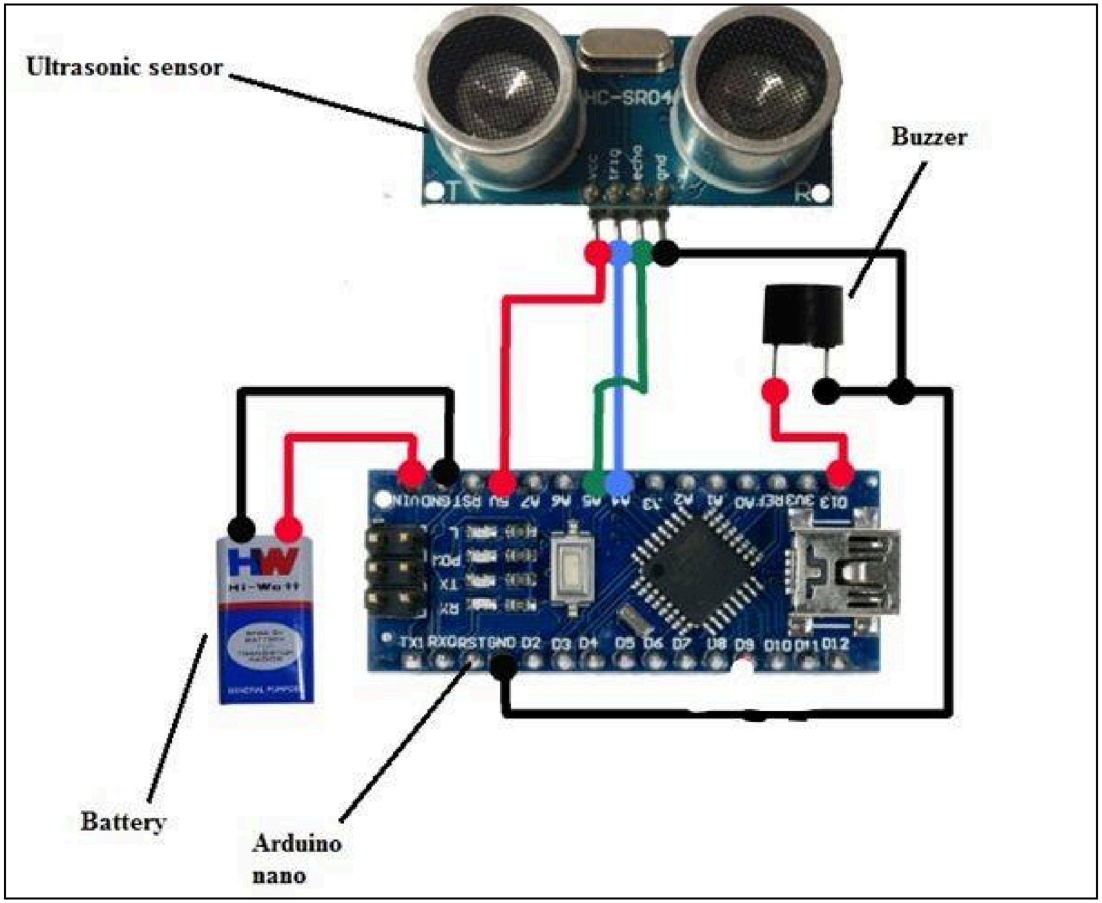
The schematic diagram provides a clear view of how the components are interconnected:

1. **Ultrasonic Sensor**:
    - **VCC**: Connected to the 5V output of the Arduino Nano.
    - **GND**: Connected to the ground (GND) pin on the Arduino.
    - **Trigger Pin**: Connected to digital pin D2 on the Arduino.
    - **Echo Pin**: Connected to digital pin D3 on the Arduino.

2. **Buzzer**:
    - **Positive Terminal**: Connected to digital pin D5 on the Arduino.
    - **Negative Terminal**: Connected to the ground (GND) pin.

3. **Push Button**:
    - **One Side**: Connected to a digital pin (D7) on the Arduino.
    - **Other Side**: Connected to ground with an internal pull-up resistor configured in the code.

4. **Power Supply (9V Battery)**:
    - **Positive Terminal**: Connected to the VIN pin on the Arduino (which has an onboard voltage regulator).
    - **Negative Terminal**: Connected to the GND pin on the Arduino.



6. **How does your invention work?**

The user wears the device as a pair of glasses. Once powered on using the push button, the ultrasonic sensor emits sound waves and calculates the time taken for the echo to return. If an object is detected within a predefined distance, the microcontroller activates the buzzer. The buzzer emits sound at 85 dB, alerting the user to the proximity of the obstacle. The system continues to monitor the surroundings, providing continuous feedback to ensure safe navigation.

7. **What are the uses of your invention and state any different ways that your invention can achieve the desired result?**

Primary use: Assisting visually impaired individuals in navigating both indoor and outdoor environments.

Customization: The threshold distance for obstacle detection can be adjusted based on the environment (e.g., reducing the distance for crowded indoor spaces).

Application in other fields: The device can be adapted for industrial use in settings where workers need to avoid unseen obstacles.

8. **Alternatives for any components/materials/steps of the invention, e.g. bolt can be replaced by rivets or adhesives.**

- **Ultrasonic Sensor** could be replaced with a **LIDAR sensor** for more precise distance measurement, though at a higher cost.
- **Buzzer** could be substituted with a **vibratory feedback mechanism** for users who prefer tactile alerts over sound.
- **Arduino Nano** can be replaced by other microcontrollers like **ESP8266** or **ESP32** for additional functionality, such as wireless connectivity.
- **Power source**: Instead of a 9V battery, a **rechargeable lithium-ion battery** could be used to extend operational life.

9. Claims

- A wearable obstacle detection system comprising an ultrasonic sensor for detecting objects in proximity, connected to a microcontroller that processes sensor data and triggers a sound-based feedback mechanism.
- A feedback system that includes a buzzer providing auditory alerts to users when an object is detected within a specified range.
- A low-power design optimized for long-term use, powered by a 9V battery, with all components mounted onto a pair of glasses.
- The use of a single push button for activating or deactivating the device, simplifying operation for visually impaired users.
- The ability to adjust detection sensitivity based on user preferences or environmental conditions.